

10

HASHING

10.1 Introduction

- Searching technique.
- Mapping keys into hash table using a hash function.
- Efficiency based on hash function used for mapping.

10.1.1 Terminology

- (i) Collision : When two keys mapped to same location, collision occurs.

$H(k_1) = H(k_2)$, where $H(k)$ is the hash function used.

- (ii) Load factor $\lambda = \frac{\text{no. of keys}}{\text{Hash table size}}$

10.1.2 Collision resolution technique

- (i) Open addressing (closed hashing)

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|-----------------------|--------------------|
| (a) Linear probing | } $\lambda \leq 1$ |
| (b) Quadratic probing | |
| (c) Double hashing | |

- (ii) Separate chaining

10.1.3 Linear Probing

Whenever there is a collision at memory location 'L', then we will search for empty slot sequentially starting from 'L' then $L + 1, L + 2, L + 3, \dots$

10.1.4 Problem with linear Probing

Linear probing suffers with primary clustering problem, consecutive keys forms a cluster and the time to find a free slot increases.

10.1.5 Quadratic Probing

Hash function $h(x) = x \bmod m$. It leads to a collision and it is i^{th} collision for key x then the collision resolution function is given as:

$$H(x, i) = (h(x) + i^2) \bmod m$$

Searching order : $L, L + 1, L + 4, L + 9, \dots$

- Free from primary clustering
- Suffers with secondary clustering problem.

10.1.6 Double Hashing : Using 2 hash functions

$(\text{hash} - 1 (\text{key}) + i * \text{hash-2} (\text{key})) \% m$

M : table size

Hash-2 (key) : Secondary hash function must not give output 0.

- * One of the best probing.
- * Uniform distribution.
- * Free from primary and secondary clustering.
- * Computation overhead because of two hash function computation.

10.2 Chaining

Uses the concept of linked list (chain) when more than one element are hashed to same location (slot) then elements are inserted into a singly linked list (chain).

- Deletion is easy compared to open addressing.
- Deleting in open addressing require rehashing remaining keys.

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